

FREEDOM INTERNATIONAL SCHOOL
I TERM ASSESSMENT – AUGUST 2026
PHYSICS – ANSWER KEY

CLASS : XII

MAX MARKS: 70
TIME: 3 HOURS

SECTION A		
1.	(A) $4 \mu\text{C}$ $q = \frac{\tau}{2a.E \sin \theta}$ $q = \frac{8 \times 10^{-3}}{0.02 \times 2 \times 10^5 \times \sin 30^\circ}$ $= 400 \times 10^{-3} \times 10^{-5}$ $q = 4 \mu\text{C}$	1
2.	(B) zero	1
3.	(B) $1.6 \times 10^{-18} \text{ J}$ $W = q(V_2 - V_1)$ $W = (1.6 \times 10^{-19}) [5 - (-5)]$ $W = 1.6 \times 10^{-18} \text{ J}$	1
4.	(B) -200 V $V_B - V_A = -Ed = -(50)(4) = -200 \text{ V}$	1
5.	(B)	1
6.	(C) 3.6 W $\text{Power, } P = \frac{V^2}{R}$ $P = \frac{(6)^2}{10}$ $P = 3.6 \text{ watt}$	1
7.	(D) 1Ω	1

	$r = \frac{E - V}{I}$ $r = \frac{8V - 4V}{4A} = \frac{4V}{4A} = 1\Omega$	
8.	(A)	1
9.	(B) remain at rest	1
10.	You are required to design an air-filled solenoid of inductance 0.016 H having a length 0.81 m and radius 0.02 m. The number of turns in the solenoid should be (A) 2592 (B) 2866 (C) 2976 (D) 3140 (B) 2866 $N^2 = \frac{Ll}{\mu_0 A}$ $N = \sqrt{\frac{0.016}{(4\pi \times 10^{-7}) \times (\pi \times (0.02)^2)} \times 0.81} = \sqrt{\frac{16 \times 81}{100 \times 1000 \times 10^{-11} \times 16\pi^2}}$ $N = \frac{4 \times 9 \times 1000}{4\pi} = \frac{9000}{\pi}$ $= N \approx 2866$	1
11.	(D) 69 V $\varepsilon = \frac{d\phi}{dt} = 16t + 5$ $\left. \frac{d\phi}{dt} \right _{t=4} = 16(4) + 5 = 64 + 5 = 69 \text{ V}$	1
12.	(D) acceleration of the magnet remains uniform as it moves through the loop.	1
For Questions 13 to 16, two statements are given one labelled Assertion (A) and other labelled Reason (R). Select the correct answer to these questions from the options as given below. A. Assertion and Reason are true and Reason is the correct explanation of Assertion. B. Assertion and Reason are true but Reason is NOT the correct explanation of Assertion. C. Assertion is true but Reason is false. D. Both Assertion and Reason are false.		
13.	A	1
14.	D	1
15.	D	1
16.	C	1
SECTION B		

17. (a)

2

$$E_x = 2 \left(k \frac{q}{a^2} \right) = \frac{2kq}{a^2}$$

(1.5)

$$E_y = 2 \left(k \frac{q}{a^2} \right) = \frac{2kq}{a^2}$$

$$E = \sqrt{E_x^2 + E_y^2} = \sqrt{\left(\frac{2kq}{a^2} \right)^2 + \left(\frac{2kq}{a^2} \right)^2} = \frac{2\sqrt{2} kq}{a^2}$$

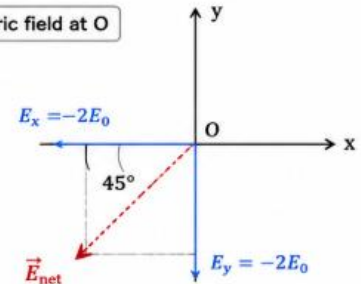
$$\tan \theta = \frac{|E_y|}{E_x}$$

$$\tan \theta = \frac{\frac{2kq}{a^2}}{\frac{2kq}{a^2}} = 1$$

(0.5)

$$\theta = \tan^{-1}(1) = 45^\circ$$

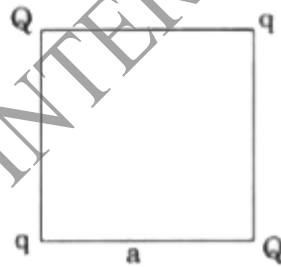
Net electric field at O



$\vec{E}$ points in the direction making 45° with the negative x-axis towards negative y-axis.

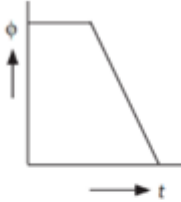
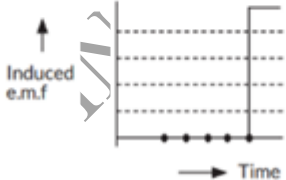
OR

(b)

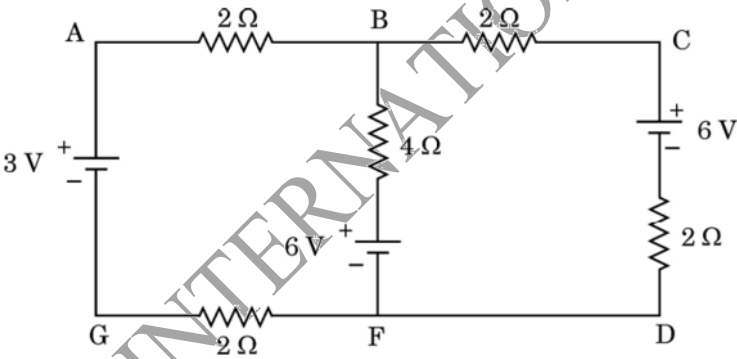


	(1) $F_1 = \frac{1}{4\pi\epsilon_0} \frac{Q^2}{(a\sqrt{2})^2} = \frac{1}{4\pi\epsilon_0} \left(\frac{Q^2}{2a^2} \right) \text{ (along AC)}$ Force due to the charge q (at B), F_2 $= \frac{1}{4\pi\epsilon_0} \frac{qQ}{a^2} \text{ along BC}$ Force due to the charge q (at D), F_3 $= \frac{1}{4\pi\epsilon_0} \frac{qQ}{a^2} \text{ along DC}$ Resultant of these two equal forces (1) $F_{23} = \frac{1}{4\pi\epsilon_0} \frac{qQ(\sqrt{2})}{a^2} \text{ (along AC)}$ $\therefore$ Net force on charge Q (at point C) $F = F_1 + F_{23} = \frac{1}{4\pi\epsilon_0} \frac{Q}{a^2} \left[\frac{Q}{2} + \sqrt{2}q \right]$ This force is directed along AC	
18.	(i) Diagram (1) (ii) No. (0.5) If the electric field were tangential, it would do work in moving a charge along the surface, changing its potential. This is not possible on an equipotential surface. (0.5)	2
19.	(i) Mobility of electron = $\frac{v_d}{E} = \frac{e\tau}{m}$ Mobility remains same, as it is independent of V.	2

	(ii) $J = nev_d$ $J = ne \cdot \frac{eV\tau}{ml}$ $J = \frac{ne^2V\tau}{ml} \quad (1)$ $J \propto V$ When V become 2V current density will become twice.	
20.	A small magnetised needle P is placed at the origin of x-y plane with its magnetic moment pointing along the y-axis. Another identical magnetised needle Q is placed in two positions, one by one. Case 1: at (a,0) with its magnetic moment pointing along x-axis. Case 2: at (0, a) with its magnetic moment pointing along y-axis. (i) In which case is the potential energy of P and Q minimum? (ii) In which case P and Q not in equilibrium? Justify your answers. (i) Case 1 $B = \frac{\mu_0 m}{4\pi a^3}$ $U = -mB \cos \theta$ $= -mB \cos 90^\circ$ $= -mB(0)$ $= 0$ Case 2 $B = \frac{\mu_0 2m}{4\pi a^3}$ $U = -mB \cos \theta$ $= -mB \cos 0^\circ$ $= -mB(1)$ $= -mB$ Potential energy is minimum in case 2 (1)	2

	(ii) Case 1 $B = \frac{\mu_0 m}{4\pi a^3}$ $\tau = mB \sin \theta$ $= mB \sin 90^\circ$ $= mB(1)$ $= mB \neq 0 \quad (1)$ Case 2 $B = \frac{\mu_0 2m}{4\pi a^3}$ $\tau = mB \sin \theta$ $= mB \sin 0^\circ$ $= mB(0)$ $= 0$ P and Q not in equilibrium in case 1	
21.	(i)  (1) (ii)  (1)	2
SECTION C		
22.	Derivation (2) Graph (1)	3
23.	Yes. The net electric flux through a closed surface can be negative if the net charge enclosed is negative. (0.5+0.5)	3

	(i) (a) For $x < r_1$: The only charge enclosed is the point charge at the center (+2q). $\Phi = \frac{2q}{\epsilon_0} \quad (0.5)$ (b) For $r_1 < x < r_2$: The enclosed charge is the point charge (2q) plus the total charge on the inner sphere (Q). $\Phi = \frac{Q + 2q}{\epsilon_0} \quad (0.5)$ (ii) (a) Inner surface of sphere (r_1): To make the field zero within the conducting wall of the sphere, the charge on its inner surface must exactly neutralize the point charge 2q at the centre. Induced charge is -2q. $\sigma_{r_1} = \frac{-2q}{4\pi r_1^2}$ $\sigma_{r_1} = -\frac{q}{2\pi r_1^2}$ (b) Inner surface of shell (r_2): To make the field zero within the conducting shell, the charge on its inner surface must neutralize the total charge inside it (the point charge 2q and the total charge Q on the inner sphere). Induced charge is $-(Q + 2q)$. $\sigma_{r_2} = \frac{-(Q + 2q)}{4\pi r_2^2}$	
24.	(i) $V_f = \frac{\text{Total charge}}{\text{Total capacitance}} = \frac{CV}{C + 2C} = \frac{CV}{3C} = \frac{V}{3}$ $Q_A = C \times \frac{V}{3} = \frac{CV}{3}$ $Q_B = 2C \times \frac{V}{3} = \frac{2CV}{3}$ $Q_A : Q_B = \frac{CV}{3} : \frac{2CV}{3} = 1 : 2$ $Q_A : Q_B = 1 : 2$	1.5 3

	(ii) $U_i = \frac{1}{2} CV^2$ $U_f = \frac{1}{2} (C + 2C) \left(\frac{V}{3} \right)^2$ $U_f = \frac{1}{2} (3C) \left(\frac{V^2}{9} \right) = \frac{CV^2}{6}$ $U_f : U_i = \frac{CV^2}{6} : \frac{CV^2}{2} = 1 : 3$ $U_f : U_i = 1 : 3$	(1.5)	
25.	 Branch AG: 0.5 A flowing from A to G Branch BF: 0.25 A flowing from F to B Branch CD: 0.25 A flowing from D to C		3
26.	Two current-carrying conductors exert a force on each other because the magnetic field produced by one conductor exerts a Lorentz force on the moving charges (current) of the other conductor. (0.5) Derivation with diagram (2) Nature of force- repulsive (0.5)		3
27.	Principle (1) Any two reasons (0.5 + 0.5) Any two factors (0.5 + 0.5)		3
28.	Definition of mutual inductance (1) Derivation (2) OR		3

	(i) Statement (0.5) Explanation (1) (ii) $\epsilon = \frac{1}{2} B \omega L^2$ (1.5) Given Values $B = 2\text{T}$ Frequency $f = 60\text{ rev/s}$ Length from centre to end $L = 1\text{m}$ $\omega = 2\pi f = 2\pi \times 60 = 120\pi\text{ rad/s}$ $\epsilon = \frac{1}{2} \times 2 \times 120\pi \times (1)^2 = 120\pi\text{V}$ $\epsilon \approx 120 \times 3.1416$ $= 376.99\text{ V}$ $\epsilon = 377\text{ V}$	
SECTION D Question numbers 29 and 30 are case study based questions. Read the following paragraphs and answer the questions that follow		
29.	(B) 2 r $E = \frac{\lambda}{2\pi\epsilon_0 r}$ $\frac{E_P}{E_Q} = \frac{r_Q}{r_P}$ $E_P = 2E_Q, \quad r_P = r$ $\frac{2E_Q}{E_Q} = \frac{r_Q}{r}$ $2 = \frac{r_Q}{r}$ $r_Q = 2r$ (II) $-7.5 \times 10^{-9}\text{ C}$	4

$$Q = \frac{Er^2}{9 \times 10^9}$$

$$Q = \frac{(3 \times 10^3)(0.15)^2}{9 \times 10^9}$$

$$Q = \frac{(3 \times 10^3)(0.0225)}{9 \times 10^9}$$

$$Q = \frac{67.5}{9 \times 10^9} = 7.5 \times 10^{-9} \text{ C}$$

OR

(B) $\frac{4}{3} \left(\frac{\lambda - 3\sigma}{\lambda - 4\sigma} \right)$

$$\frac{E_A}{E_B} = \frac{\frac{\lambda}{2\pi\epsilon_0 \left(\frac{3}{\pi} \right)} - \frac{\sigma}{2\epsilon_0}}{\frac{\lambda}{2\pi\epsilon_0 \left(\frac{4}{\pi} \right)} - \frac{\sigma}{2\epsilon_0}}$$

$$= \frac{4}{3} \left(\frac{\lambda - 3\sigma}{\lambda - 4\sigma} \right)$$

(III) (B) the field at B is $\frac{Q}{A\epsilon_0}$

(IV) C. half

$$E = \frac{Q}{\epsilon_0 A}$$

$$A' = 2A$$

$$E' = \frac{Q}{\epsilon_0 (2A)}$$

$$E' = \frac{1}{2} \left(\frac{Q}{\epsilon_0 A} \right)$$

30. (A) repelled by both the poles
 (II) (A) aluminium
 (III) (C) less than that of free space
 (D)

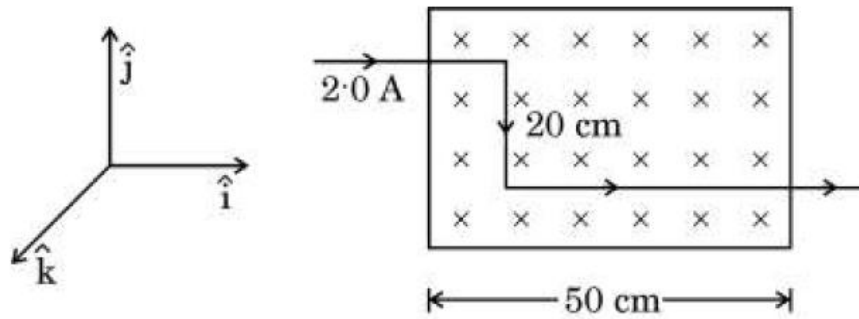
OR

4

	(IV) (B) A ferromagnetic substance changes into a paramagnetic substance above the Curie temperature	
SECTION E		
31.	(i) Graph (1) (ii) Derivation (2) (iii) (2) $U = k \left(\frac{q_1 q_2}{r_{12}} + \frac{q_2 q_3}{r_{23}} + \frac{q_3 q_1}{r_{31}} \right)$ $q_1 = +q \text{ at } (3a, 0)$ $q_2 = -q \text{ at } (0, 4a)$ $q_3 = Q \text{ at } (0, 0)$ $r_{12} = \sqrt{(3a)^2 + (4a)^2} = 5a, \quad r_{23} = 4a, \quad r_{31} = 3a$ $0 = k \left(\frac{(+q)(-q)}{5a} + \frac{(-q)Q}{4a} + \frac{Q(+q)}{3a} \right)$ $0 = \frac{k}{a} \left(-\frac{q^2}{5} - \frac{qQ}{4} + \frac{qQ}{3} \right)$ $-\frac{q^2}{5} + qQ \left(\frac{1}{3} - \frac{1}{4} \right) = 0$ $-\frac{q^2}{5} + \frac{qQ}{12} = 0$ $\frac{qQ}{12} = \frac{q^2}{5}$ $Q = \frac{12q}{5}$ OR (i) Derivation (3)	5

	(ii) $V = \frac{kQ}{R}$ $V = \frac{9 \times 10^9 \times 6 \times 10^{-6}}{0.2} \quad (1)$ $V = \frac{54 \times 10^3}{0.2}$ $V = 2.7 \times 10^5 \text{ V}$ (i) Potential at the surface: $V = 2.7 \times 10^5 \text{ V} \quad (0.5)$ (ii) Potential at the centre: $V = 2.7 \times 10^5 \text{ V} \quad (0.5)$	
32.	(i) Derivation with proper diagram (3) Diagram (0.5) (ii) $I = \frac{e}{T} = \frac{ev}{2\pi r} \quad (1.5)$ $I = \frac{(1.6 \times 10^{-19})(10^7)}{2\pi(10^{-10})}$ $I = \frac{1.6 \times 10^{-12}}{6.28 \times 10^{-10}}$ $I = 2.55 \times 10^{-3} \text{ A}$ $I = 2.55 \times 10^{-3} \text{ A}$ OR (i) Derivation (2) Force is maximum, when length of conductor is perpendicular to magnetic field. (0.5)	5

(ii)



$$\vec{F}_{\text{net}} = I(\vec{L}_{\text{net}} \times \vec{B})$$

Current (I): 2.0 A

Magnetic Field ($\vec{B}$): $-(0.50 \text{ T})\hat{k}$ (directed into the page)

Net Horizontal Displacement (Δx): +50 cm = +0.50 m (along $\hat{i}$)

Net Vertical Displacement (Δy): -20 cm = -0.20 m (along $\hat{j}$)

$$\vec{L}_{\text{net}} = 0.50\hat{i} - 0.20\hat{j} \text{ m}$$

$$\vec{F}_{\text{net}} = 2.0 [(0.50\hat{i} - 0.20\hat{j}) \times (-0.50\hat{k})]$$

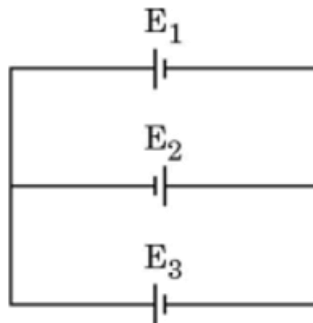
$$\vec{F}_{\text{net}} = 2.0 [0.10\hat{i} + 0.25\hat{j}] = 0.20\hat{i} + 0.50\hat{j} \text{ N}$$

$$F_{\text{net}} = \sqrt{(0.20)^2 + (0.50)^2}$$

$$0.54 \text{ N}$$

(2.5)

33. (a) (i)



5

$$E_{eq} = \frac{\frac{E_1}{r_1} - \frac{E_2}{r_2} + \frac{E_3}{r_3}}{\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}}$$

$$E_{eq} = \frac{\frac{4}{2} - \frac{2}{4} + \frac{6}{2}}{\frac{1}{2} + \frac{1}{4} + \frac{1}{2}} = \frac{2 - 0.5 + 3}{1.25} = \frac{4.5}{1.25} = 3.6 \text{ V}$$

Since there is no external load resistance in the outer circuit, the terminal potential difference across the parallel combination is equal to the equivalent EMF:

$$V = 3.6 \text{ V}$$

For battery E_1 :

$$I_1 = \frac{E_1 - V}{r_1} = \frac{4 - 3.6}{2} = 0.2 \text{ A} \quad (1)$$

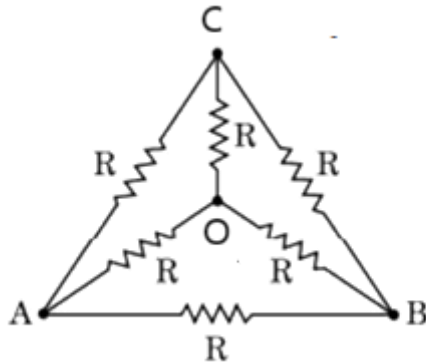
For battery E_2 (reversed branch):

$$I_2 = \frac{-E_2 - V}{r_2} = \frac{-2 - 3.6}{4} = \frac{-5.6}{4} = -1.4 \text{ A}$$

For battery E_3 :

$$I_3 = \frac{E_3 - V}{r_3} = \frac{6 - 3.6}{2} = 1.2 \text{ A} \quad (1)$$

(ii)



Since the bridge is balanced, the current through the bridge arm R_{CO} is zero, so R_{CO} is removed.

The remaining network consists of three parallel branches across terminals A and B:

Upper branch ($R_{AC} + R_{CB}$ in series): $R_1 = R + R = 2R$

Lower branch ($R_{AO} + R_{OB}$ in series): $R_2 = R + R = 2R$

Bottom branch (direct connection): $R_3 = R$

$$\frac{1}{R_{AB}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R_{AB}} = \frac{1}{2R} + \frac{1}{2R} + \frac{1}{R}$$

$$\frac{1}{R_{AB}} = \frac{2}{2R} + \frac{1}{R} = \frac{1}{R} + \frac{1}{R} = \frac{2}{R} \quad (1)$$

$$R_{AB} = \frac{R}{2}$$

OR

(b) (i) Showing drift velocity independent of time (1)

Derivation (2)

(ii) When a circuit is closed, the electric field is established throughout the conductor almost instantaneously. As a result, all the free electrons in the conductor begin to drift simultaneously under the influence of the electric field. Therefore, the current is established almost immediately throughout the circuit. (0.5)

(iii)

Since both wires are copper,

$$n_1 = n_2 = n.$$

$$e_1 = e_2 = e$$

$I_1 = I_2 = I$ (same current flows through series circuit).

(1.5)

$$v_d = \frac{I}{neA}$$

$$\frac{v_{d1}}{v_{d2}} = \frac{A_2}{A_1}$$

$$= \frac{\pi r_2^2}{\pi r_1^2} = \frac{2^2}{3^2} = \frac{4}{9}$$

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